Student record management

Course title : Structured Programming

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Section: 25

Presented by :

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INTRODUCTION

The Student Record Management System is a console-based application designed to help educational institutions manage student information efficiently. This project involves creating, storing, updating, and retrieving student records, which include details like student ID, name, age, gender, and grades.

Student record management

Code

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
// Define a structure to store student information
typedef struct {
    char name[50];
    int rollNumber;
    int totalMarks;
} Student;
// Define a structure to manage the list of students
typedef struct {
    Student students[100];
    int count;
} StudentDatabase;

 StudentDatabase database = { .count = 0 };
```

Student record management

Code

Display the menu

```
void displayMenu() {

    printf("Menu:\n");
    printf("1. Add Student Record\n");
    printf("2. Delete Student Record\n");
    printf("3. Update Student Record\n");
    printf("4. Display All Records\n");
    printf("5. Display Specific Record\n");
    printf("6. Exit\n");
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice:
```

Code

Add a student record

```
void addStudentRecord() {
    Student student;

    printf("Enter student name: ");
    scanf("%s", student.name);
    printf("Enter roll number: ");
    scanf("%d", &student.rollNumber);
    printf("Enter total marks: ");
    scanf("%d", &student.totalMarks);

    database.students[database.count++] = student;
    printf("Student record added successfully.\n");
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 1
Enter student name: zahedul
Enter roll number: 1
Enter total marks: 100
Student record added successfully.
Do you want to add more records? (y/n):
```

Student record management

Code

Student record by roll number

```
void deleteStudentRecord() {
    int rollNumber;
    printf("Enter roll number of student to delete: ");
    scanf("%d", &rollNumber);
    int found = 0;
    for (int i = 0; i < database.count; i++) {
        if (database.students[i].rollNumber == rollNumber) {
            found = 1;
            for (int j = i; j < database.count - 1; j++) {
                database.students[j] = database.students[j + 1];
            }
            database.count--;
            printf("Student record deleted successfully.\n");
            break;        }
    }    if (!found) {
        printf("Student record not found.\n");    }
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 2
Enter roll number of student to delete: 3
Student record deleted successfully.
Do you want to return to the menu? (y/n):
```

Code

Student upadate record by roll number

```
void updateStudentRecord() {
    int rollNumber;
    printf("Enter roll number of student to update: ");
    scanf("%d", &rollNumber);
    int found = 0;
    for (int i = 0; i < database.count; i++) {
        if (database.students[i].rollNumber == rollNumber) {
            found = 1;
            printf("Enter new total marks: ");
            scanf("%d", &database.students[i].totalMarks);
            printf("Student record updated successfully.\n");
            break;        }
    }
    if (!found) {
        printf("Student record not found.\n");
    }
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 3
Enter roll number of student to update: 2
Enter new total marks: 89
Student record updated successfully.
Do you want to return to the menu? (y/n):
```

Code

Display all student records

```
void displayAllRecords() {
    if (database.count == 0) {
        printf("No records found.\n");
        return;
    }

    for (int i = 0; i < database.count; i++) {
        printf("Name: %s, Roll Number: %d, Total Marks: %d\n",
               database.students[i].name,
database.students[i].rollNumber, database.students[i].totalMarks);
    }
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 4
Name: zahedul, Roll Number: 1, Total Marks: 100
Name: Adri, Roll Number: 2, Total Marks: 89
Do you want to return to the menu? (y/n):
```

Student record management

Code

Specific student record by roll number

```
void displaySpecificRecord() {
    int rollNumber;
    printf("Enter roll number of student to display: ");
    scanf("%d", &rollNumber);
    int found = 0;
    for (int i = 0; i < database.count; i++) {
        if (database.students[i].rollNumber == rollNumber) {
            found = 1;
            printf("Name: %s, Roll Number: %d, Total Marks: %d\n",
                   database.students[i].name, database.students[i].rollNumber,
database.students[i].totalMarks);
            break;        }
    }
    if (!found) {
        printf("Student record not found.\n");    }
}
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 5
Enter roll number of student to display: 2
Name: Adri, Roll Number: 2, Total Marks: 89
Do you want to return to the menu? (y/n):
```

Student record management

Code

```
int main() {
    int choice;
    while (1) {
        displayMenu();
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                addStudentRecord();
                char addMore;
                do {
                    printf("Do you want to add more records? (y/n): ");
                    scanf(" %c", &addMore);
                    if (addMore == 'y' || addMore == 'Y') {
                        addStudentRecord();                    }
                }
                while (addMore == 'y' || addMore == 'Y');
                break;
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 1
Enter student name: zahedul
Enter roll number: 1
Enter total marks: 100
Student record added successfully.
Do you want to add more records? (y/n): y
Enter student name: Adri
Enter roll number: 2
Enter total marks: 99
Student record added successfully.
Do you want to add more records? (y/n): |
```

Student record management

Code

```
        case 2:
            deleteStudentRecord();
            break;
        case 3:
            updateStudentRecord();
            break;
        case 4:
            displayAllRecords();
            break;
        case 5:
            displaySpecificRecord();
            break;
        case 6:
            exit(0);
        default:
            printf("Invalid choice. Please try again.\n");
    }
```

```
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
Enter your choice: 6

Process returned 0 (0x0)   execution time : 922.728 s
Press any key to continue.
```

Student record management

Code

```
char openMenu;
        printf("Do you want to return to the menu? (y/n): ");
        scanf(" %c", &openMenu);
        if (openMenu != 'y' && openMenu != 'Y') {
            break;
        }
    }

    return 0;
}
```

```
Enter student name: diya
Enter roll number: 3
Enter total marks: 67
Student record added successfully.
Do you want to add more records? (y/n): n
Do you want to return to the menu? (y/n): y
Menu:
1. Add Student Record
2. Delete Student Record
3. Update Student Record
4. Display All Records
5. Display Specific Record
6. Exit
```

OVERVIEW

1. A console-based system for managing student records, supporting operations like adding, deleting, updating, and viewing records.

2. Uses C structures (`Student` and `StudentDatabase`) for efficient data storage and modular functions for clarity.

3. Features a user-friendly, menu-driven interface with options for iterative actions and easy navigation.

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THANK YOU

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